Machine learning for data analytics

Assignment 7: Federated Learning & MLOps

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Each assignment should be submitted as a Jupyter notebook with complete code, visualizations, and written analysis.

Federated Learning (40 points)

Implement federated learning using Flower:

- Simulate 5 clients with non-IID data splits (MNIST).
- Implement FedAvg algorithm.
- Track model performance at each communication round.
- Compare with centralized training.

Differential Privacy (30 points)

Apply differential privacy to neural network training:

- Use Opacus library with PyTorch.
- Train on sensitive dataset (e.g., Adult Income).
- Experiment with different epsilon values (1.0, 5.0, 10.0).
- Analyze privacy-utility tradeoff.

MLOps Pipeline (30 points)

Build complete ML pipeline:

- Track experiments using MLflow.
- Version data with DVC.
- Deploy model as REST API using FastAPI.
- Containerize with Docker.
- Implement basic monitoring for predictions.